

RUBINA REKIĆ

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EDUCATION

University of Sarajevo

2023 - 2026

Bachelor of Engineering in Computer Science

Sarajevo, Bosnia and Herzegovina

GPA: 8.8/10.0

Thesis: "Application of Graph Theory to Neural Network Modelling"

JU "Gimnazija" Cazin

2019 – 2023

High School Diploma

Cazin, Bosnia and Herzegovina

Valedictorian

SKILLS

Programming: C#, Kotlin, Java, JavaScript, Python, C++, C, SQL, Git, LaTeX

Web & Frameworks: React, Node.js, Express.js, ASP.NET Core, Flask, Tailwind CSS

Mobile: Android Studio, REST API integration

Other: MQTT, OOP, Design Patterns, TDD

Languages: Bosnian (native) | English (professional) | German (limited working proficiency)

EXPERIENCE AND PROJECTS

Encode d.o.o.

March 2025 – Present

Mathematics Instructor

Delivered preparatory classes for university students from the Faculty of Mechanical and Electrical Engineering

Application of Graph Theory to Neural Network Modelling - Bachelor's Thesis

2026

Formalized trained feedforward neural networks as directed weighted graphs and applied centrality, modularity (Louvain), spectral (Laplacian), and robustness analysis to compare three architectures of varying depth and width

ScenarijPro - Full-Stack Web Application

2024/2025

Built a collaborative screenplay editor with real-time conflict resolution, version control checkpoints, and script analysis

Developed iteratively across 4 sprints: from static UI to a full Node.js/Express REST API with MySQL and Sequelize ORM

YourRide - Full-Stack Web Application

Spring 2025

Developed a ride-sharing platform using ASP.NET Core with SignalR for real-time notifications and role-based access control

Designed full system architecture including UML diagrams, applied SOLID principles and structural design patterns

FilmMate QA - .NET Application

Spring 2025

Applied TDD (Red-Green-Refactor), MSTest unit testing with 90%+ coverage, and white/black-box testing techniques

NewsFeed Android App - Mobile Application

Spring 2025

Developed an Android application in Kotlin with real-time news fetching via REST API and local persistence using Room database

Implemented category filtering, sorting, and automatic feed refresh with 3-second polling interval

Modeled biological neural networks as directed graphs using graph theory (Dale's Law, Barabási-Albert model) and analyzed their structure and robustness

Laid the groundwork for my Bachelor's thesis, which extended this graph-theoretic approach to ANN